

Multilayer Graph Learning for Energy-Efficient User Association and Cell Switch-Off in Dense Wireless Networks

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Abstract—Dense 5G and beyond-5G radio access networks require user association policies that reduce infrastructure energy consumption while preserving quality-of-service (QoS) and load feasibility. Conventional received-signal-strength, SINR, and load-aware association rules are computationally simple but do not explicitly optimize the active base-station (BS) set. Meanwhile, many graph-learning approaches represent the wireless system by a single aggregated graph, which can obscure heterogeneous relations that are relevant to energy-aware control. This paper proposes a multilayer graph-learning framework for joint user association and BS switch-off. Each wireless snapshot is modeled as a UE-level multilayer graph with separate relation layers for association competition, interference similarity, load-pressure similarity, and temporal traffic similarity. Fixed-fusion and attention-weighted multilayer graph convolutional networks (ML-GCNs) learn UE-to-BS association preferences, while an active-cell selector and a QoS/load repair procedure convert the learned preferences into feasible energy-aware decisions. The framework is analyzed in terms of permutation equivariance, aggregation-induced information loss, and feasibility-preserving repair. Simulation results show that the proposed ML-GCN achieves 55.2% mean energy saving relative to an all-on network while maintaining a served ratio of 1.0 and using 3.18 active BSs on average. In contrast, RSRP, SINR, and load-aware baselines save only about 17–18% energy because they do not compact the active-cell set. Ablation and sensitivity studies show that preserving layer-specific wireless relations improves active-BS compactness and robustness across bandwidth, QoS, UE-density, and shadowing variations. Runtime results further indicate that ML-GCN provides a practical feed-forward approximation to finite active-cell search.

Index Terms—Graph neural networks, multilayer networks, user association, network energy savings, 6G, base-station switch-off.

I. INTRODUCTION

DENSE radio access networks are expected to support rapidly increasing traffic demand while reducing the energy footprint of wireless infrastructure. This tension is especially visible in ultra-dense 5G and beyond-5G deployments, where many base stations (BSs) are installed to improve coverage, capacity, and reliability, but only a subset of them may be necessary under a given traffic realization. Since a BS consumes non-negligible static power even when lightly loaded, energy-efficient network operation requires more than selecting the strongest serving cell for each user equipment (UE). It requires a joint decision: which BSs should remain active, which UEs should be associated with them, and how these decisions can preserve service constraints under interference, load, and traffic variability.

User association has therefore become a central control problem in dense wireless networks. Classical association rules based on received signal strength, signal-to-interference-plus-noise ratio (SINR), or load balancing are computationally attractive, but they are fundamentally local. They typically assign each UE using the quality or load of a small number of candidate BSs and do not explicitly reason about the global energy consequence of activating or deactivating BSs. Optimization-based formulations can model the joint association and switch-off problem more directly, but the resulting mixed discrete-continuous decision space is difficult to solve repeatedly at network time scales. Learning-based methods are a natural alternative because they can amortize computation after training; however, the quality of a learned policy depends strongly on the graph representation through which the wireless system is presented to the model.

Graph neural networks (GNNs) are attractive for wireless resource management because wireless networks are relational systems. A UE is not an isolated sample: its best association depends on neighboring UEs, competing BSs, interference coupling, traffic demand, and load created by other association decisions. Recent wireless GNN studies show that graph-based models can provide scalable approximations for resource-allocation tasks and can exploit permutation structure in wireless graphs [1], [2]. Nevertheless, many graph-learning formulations reduce the system to a single relation graph. Such a collapse is convenient, but it can hide an important structural distinction: a dense wireless snapshot contains multiple kinds of relations whose semantics are not interchangeable. Two UEs may compete for the same serving BS, exhibit similar traffic demand, experience correlated interference conditions, or follow similar temporal traffic patterns. These relations may overlap, but they are not equivalent and should not necessarily be represented by a single averaged graph.

This observation motivates a multilayer network formulation. In multilayer network models, a system is represented not only by nodes and edges, but also by layers that encode different interaction types, time slices, modalities, or couplings [3]–[5]. This distinction is not only descriptive. Prior work on multilayer networks has shown that aggregation can remove structural information, distort centrality rankings, and obscure layer-specific organization [6], [7]. For energy-aware wireless control, this issue is consequential. If association competition, interference, load pressure, and temporal traffic similarity are collapsed into one graph, a learning model may receive an averaged relation that no longer identifies which physical or

operational mechanism caused a dependency between UEs. A policy trained on such an aggregate graph may therefore associate users correctly in easy cases but fail to learn why certain BSs can be switched off without violating quality-of-service (QoS) and load constraints.

In this paper, we study energy-efficient user association and BS switch-off as a multilayer graph decision problem. For each wireless snapshot, we construct a UE-level multilayer graph

$$\mathcal{G} = (\mathcal{V}, \mathbf{X}, \{\mathbf{A}^{(\ell)}\}_{\ell \in \mathcal{L}}), \quad (1)$$

where $\mathcal{V}$ is the set of UEs, $\mathbf{X}$ contains UE radio and traffic features, and each layer $\mathbf{A}^{(\ell)}$ represents a distinct wireless relation. In the present study, the layer set $\mathcal{L}$ consists of association competition, interference similarity, load-pressure similarity, and temporal traffic similarity. A layer-aware GNN then learns separate graph filters over these relation layers before fusing the resulting embeddings for association and active-cell decisions. The final decision is evaluated by network energy, QoS satisfaction, BS activity, and load feasibility rather than by assignment accuracy alone.

The central research question is:

Can preserving multilayer wireless structure improve energy-aware user association and BS switch-off decisions, compared with conventional association heuristics and single-layer graph-learning models, while maintaining QoS and load feasibility?

The proposed framework has two linked components. First, a multilayer GNN predicts UE-to-BS association preferences using layer-specific graph propagation. Second, an active-cell selector uses the learned association distribution and BS-level radio/load summaries to identify a compact active BS set. A QoS- and load-aware repair procedure then converts the learned preferences into a feasible association whenever possible, activating additional BSs only when the predicted active set cannot support the traffic realization. This design separates representation learning from constraint enforcement: the GNN learns multilayer structure, while the repair stage protects wireless feasibility.

The paper makes the following contributions.

- 1) We formulate joint user association and BS switch-off as a multilayer network decision problem, with separate layers for association competition, interference coupling, load pressure, and temporal traffic similarity. The formulation preserves layer-specific wireless dependencies and admits theoretical analysis of permutation equivariance, aggregation-induced information loss, and feasibility-preserving decision repair.
- 2) We develop fixed-fusion and attention-weighted multilayer GNN models for UE association preferences and combine them with an active-cell selector for energy-aware BS switch-off. The resulting policy is evaluated under explicit QoS, load, and energy constraints.
- 3) We define a comparative metric suite for constrained wireless control, including energy saving relative to an all-on network, energy gap relative to the oracle, served ratio, active BS count, maximum load, and mean active load. Layer-ablation gains, attention entropy, and

layer-redundancy measures are included to assess the contribution and distinctiveness of each relation layer.

- 4) We conduct a simulation-based comparative study against conventional RSRP, SINR, and load-aware association heuristics, single-layer graph learning, aggregated GCN learning, fixed multilayer GCN learning, and attention-weighted multilayer GCN learning. The experiments are designed to determine not only whether a policy saves energy, but whether it does so while preserving service feasibility.

The remainder of this paper is organized as follows. Section II reviews wireless user association, graph learning for resource management, and multilayer graph modeling. Section III presents the system model and energy-aware association objective. Section IV develops the proposed multilayer graph-learning framework and its mathematical properties. Section V describes the simulation design, baselines, and metrics. Section VI reports and discusses the comparative results. Section VII concludes the paper.

II. RELATED WORK

Energy-aware user association has been widely studied as a mechanism for improving spectral efficiency, load distribution, and infrastructure energy consumption in dense wireless networks. Conventional association rules based on received signal strength, SINR, or load provide simple and scalable decisions, but they are usually myopic with respect to BS switch-off because the activation cost of each BS is not directly optimized. Optimization-based methods can incorporate energy and service constraints explicitly, but the joint association and activation problem is combinatorial and must be solved under changing channel and traffic conditions.

GNNs provide a learning-based alternative for resource-management problems in which the wireless topology varies across network realizations. Random-edge GNNs have been used to approximate wireless resource allocation over changing interference graphs [1], while scalable GNN architectures have been analyzed for radio resource management with permutation properties and size generalization [2]. Graph attention has also been applied to wireless user association [8]. These studies support the use of graph learning for wireless control, but they typically represent the network through a single graph or a single dominant relation.

Multilayer network models provide a formal way to represent systems with multiple edge semantics, time slices, or interaction channels [3]–[5]. The importance of preserving layer structure is supported by work on structural reducibility and multilayer node ranking, where aggregation can remove relevant information or change structural rankings [6], [7]. These results are relevant to dense wireless networks because association competition, interference, load, and temporal traffic similarity describe different dependencies among UEs.

Several recent GNN studies have extended graph learning to multilayer or multi-view settings. The mGNN framework generalizes GNN operations to multilayer networks [9]. Dynamic-multilayer GCNs have been used to combine intralayer and interlayer convolution for industrial monitoring [10], while

attention-based dynamic multilayer GNNs have been developed for temporal multilayer prediction tasks [11]. MVMA-GCN further uses multi-view attention and relationship dissimilarity to learn from multiple node-relation graphs [12]. These works demonstrate the value of layer-aware graph learning, but they do not address energy-constrained wireless association with active-cell control. The present work focuses on this setting by coupling a multilayer UE representation with BS switch-off, QoS/load repair, and energy-oriented evaluation.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Network Snapshot

Consider a dense downlink cellular network with a set of BSs $\mathcal{B} = \{1, \dots, B\}$ and a set of UEs $\mathcal{U} = \{1, \dots, N\}$ observed at a traffic snapshot t . The position, traffic demand, and radio measurements of UE i are represented by a feature vector $\mathbf{x}_i \in \mathbb{R}^F$, and the snapshot feature matrix is $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]^T \in \mathbb{R}^{N \times F}$. Let $d_i(t)$ denote the traffic demand of UE i in Mbps. The large-scale and small-scale channel gain from BS b to UE i is denoted by $g_{i,b}(t)$, and the BS transmit power is P_{tx} .

The user association and BS switch-off decision is described by two binary variables. Let

$$z_{i,b} = \begin{cases} 1, & \text{if UE } i \text{ is associated with BS } b, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

and let $a_b \in \{0, 1\}$ indicate whether BS b is active. Each UE is associated with exactly one BS:

$$\sum_{b \in \mathcal{B}} z_{i,b} = 1, \quad \forall i \in \mathcal{U}, \quad (3)$$

and a UE can be assigned only to an active BS:

$$z_{i,b} \leq a_b, \quad \forall i \in \mathcal{U}, b \in \mathcal{B}. \quad (4)$$

B. SINR, Rate, and Load

For a given active-BS vector $\mathbf{a} = [a_1, \dots, a_B]^T$, the downlink SINR that UE i would obtain from BS b is

$$\gamma_{i,b}(\mathbf{a}) = \frac{a_b P_{\text{tx}} g_{i,b}}{\sigma^2 + \sum_{c \in \mathcal{B}, c \neq b} a_c P_{\text{tx}} g_{i,c}}, \quad (5)$$

where σ^2 is the receiver noise power. The corresponding achievable rate is modeled as

$$R_{i,b}(\mathbf{a}) = W \log_2 (1 + \gamma_{i,b}(\mathbf{a})), \quad (6)$$

where W is the system bandwidth. The RSRP feasibility condition is

$$P_{\text{tx}} g_{i,b} \geq \Gamma_{\text{rsrp}}, \quad (7)$$

and the minimum-rate condition is

$$R_{i,b}(\mathbf{a}) \geq R_{\text{min}}. \quad (8)$$

The load contribution of UE i to BS b is the fraction of radio resources needed to support its demand:

$$\ell_{i,b}(\mathbf{a}) = \frac{d_i}{R_{i,b}(\mathbf{a})}. \quad (9)$$

The total load of BS b under association matrix $\mathbf{Z}$ is

$$\rho_b(\mathbf{Z}, \mathbf{a}) = \sum_{i \in \mathcal{U}} z_{i,b} \ell_{i,b}(\mathbf{a}), \quad (10)$$

and the load constraint is

$$\rho_b(\mathbf{Z}, \mathbf{a}) \leq \rho_{\text{max}}, \quad \forall b \in \mathcal{B}. \quad (11)$$

C. Energy Model

The network energy model separates the static cost of keeping a BS active, the load-dependent cost of serving traffic, and the sleep-mode power of inactive BSs. The power consumed by BS b is

$$P_b(\mathbf{Z}, \mathbf{a}) = a_b (P_{\text{fix}} + P_{\text{dyn}} \rho_b) + (1 - a_b) P_{\text{sleep}}, \quad (12)$$

where P_{fix} is the fixed active power, P_{dyn} is the dynamic load-dependent power coefficient, and P_{sleep} is the sleep-mode power. The total network power is

$$P_{\text{net}}(\mathbf{Z}, \mathbf{a}) = \sum_{b \in \mathcal{B}} P_b(\mathbf{Z}, \mathbf{a}). \quad (13)$$

The all-on reference power is obtained by setting $a_b = 1$ for all $b \in \mathcal{B}$. This reference is used to report energy saving:

$$\text{ES} = \frac{P_{\text{all}} - P_{\text{net}}}{P_{\text{all}}}. \quad (14)$$

D. Optimization Problem

The ideal energy-aware association problem is to minimize network power while satisfying association, activation, QoS, and load constraints:

$$\begin{aligned} \min_{\mathbf{Z}, \mathbf{a}} \quad & P_{\text{net}}(\mathbf{Z}, \mathbf{a}) \\ \text{s.t.} \quad & \sum_{b \in \mathcal{B}} z_{i,b} = 1, \quad \forall i \in \mathcal{U}, \\ & z_{i,b} \leq a_b, \quad \forall i \in \mathcal{U}, b \in \mathcal{B}, \\ & z_{i,b} P_{\text{tx}} g_{i,b} \geq z_{i,b} \Gamma_{\text{rsrp}}, \quad \forall i \in \mathcal{U}, b \in \mathcal{B}, \\ & z_{i,b} R_{i,b}(\mathbf{a}) \geq z_{i,b} R_{\text{min}}, \quad \forall i \in \mathcal{U}, b \in \mathcal{B}, \\ & \rho_b(\mathbf{Z}, \mathbf{a}) \leq \rho_{\text{max}}, \quad \forall b \in \mathcal{B}, \\ & z_{i,b} \in \{0, 1\}, a_b \in \{0, 1\}. \end{aligned} \quad (15)$$

This problem is combinatorial because the association matrix and active-BS vector are discrete and coupled through SINR, rate, load, and power. The goal of the proposed learning framework is therefore not to solve (15) exactly at every snapshot, but to learn a scalable policy that approximates low-energy feasible decisions from the multilayer wireless graph representation.

IV. PROPOSED MULTILAYER GRAPH LEARNING FRAMEWORK

A. Overview

The proposed framework converts each wireless snapshot into a multilayer UE graph, learns UE-to-BS association preferences using layer-specific graph filters, predicts a compact active-BS set, and applies a QoS/load repair step before evaluation. Fig. 1 gives the computational pipeline, while

Fig. 2 gives the detailed multilayer model architecture. The key modeling choice is that different wireless relations are not merged before graph propagation. Instead, each relation layer is processed by a dedicated graph filter and fused only after layer-wise representation learning.

B. Multilayer Wireless Graph Construction

For each snapshot, let $\mathbf{G} \in \mathbb{R}^{N \times B}$ denote the channel-gain matrix with entries $g_{i,b}$, and let $\mathbf{r}_i$ be the position of UE i . The proposed model builds a multilayer graph

$$\mathcal{M} = (\mathcal{U}, \mathbf{X}, \{\mathbf{S}^{(\ell)}\}_{\ell \in \mathcal{L}}), \quad \mathcal{L} = \{a, i, l, t\}, \quad (16)$$

where a , i , l , and t denote association, interference, load, and temporal layers, respectively. Each layer is constructed as a nonnegative adjacency matrix $\mathbf{A}^{(\ell)}$ and converted into a row-stochastic support matrix

$$\mathbf{S}^{(\ell)} = (\mathbf{D}^{(\ell)})^{-1} (\mathbf{A}^{(\ell)} + \mathbf{I}), \quad (17)$$

where $\mathbf{D}^{(\ell)}$ is the diagonal row-sum matrix of $\mathbf{A}^{(\ell)} + \mathbf{I}$. The identity term preserves each UE's own features during propagation.

The association layer connects UEs that compete for the same strongest BS. Let

$$b_i^* = \arg \max_{b \in \mathcal{B}} g_{i,b}. \quad (18)$$

Then

$$A_{j,k}^{(a)} = \mathbf{1}\{b_j^* = b_k^*\}, \quad j \neq k. \quad (19)$$

The interference layer compares normalized channel-gain profiles. Define $\mathbf{q}_j = \mathbf{g}_j / \|\mathbf{g}_j\|_1$, where $\mathbf{g}_j$ is the j th row of $\mathbf{G}$. The interference similarity is

$$A_{j,k}^{(i)} = \mathbf{q}_j^T \mathbf{q}_k, \quad j \neq k. \quad (20)$$

The load layer links UEs with similar traffic demand within the same association region:

$$A_{j,k}^{(l)} = A_{j,k}^{(a)} \exp\left(-\eta_l \frac{|d_j - d_k|}{d_{\max}}\right), \quad (21)$$

where $\eta_l > 0$ controls demand-similarity sharpness and $d_{\max}$ is the maximum demand scale. Finally, the temporal layer captures spatial-demand similarity modulated by the traffic period:

$$A_{j,k}^{(t)} = h(t) \exp\left(-\eta_r \frac{\|\mathbf{r}_j - \mathbf{r}_k\|_2}{R}\right) \exp\left(-\eta_d \frac{|d_j - d_k|}{d_{\max}}\right), \quad (22)$$

where R is the cell-area radius and $h(t)$ is a bounded periodic traffic factor. Sparse top- K thresholding is applied to the similarity layers to avoid dense message passing and to emphasize dominant local dependencies.

C. Layer-Aware Association Learning

Given $\mathbf{X}$ and the supports $\{\mathbf{S}^{(\ell)}\}$, the fixed-fusion multilayer GCN computes one hidden representation per layer:

$$\mathbf{H}^{(\ell)} = \phi\left(\mathbf{S}^{(\ell)} \mathbf{X} \mathbf{W}^{(\ell)}\right), \quad \ell \in \mathcal{L}, \quad (23)$$

where $\phi(\cdot)$ is a pointwise nonlinearity and $\mathbf{W}^{(\ell)}$ is a layer-specific trainable matrix. The layer-wise embeddings are fused as

$$\mathbf{E} = [\mathbf{H}^{(a)} \parallel \mathbf{H}^{(i)} \parallel \mathbf{H}^{(l)} \parallel \mathbf{H}^{(t)}], \quad (24)$$

where $\parallel$ denotes feature concatenation. The UE-to-BS association probability matrix is

$$\mathbf{P} = \text{softmax}(\mathbf{E} \mathbf{W}_o + \mathbf{b}_o), \quad (25)$$

where $P_{i,b}$ is the learned probability that UE i should be associated with BS b .

The attention-weighted model assigns a trainable nonnegative importance weight to each layer. Let η_ℓ be the attention logit of layer ℓ . The normalized layer weight is

$$\alpha_\ell = \frac{\exp(\eta_\ell)}{\sum_{r \in \mathcal{L}} \exp(\eta_r)}. \quad (26)$$

The fused embedding becomes

$$\mathbf{E}_{\text{att}} = \left[|\mathcal{L}| \alpha_a \mathbf{H}^{(a)} \parallel |\mathcal{L}| \alpha_i \mathbf{H}^{(i)} \parallel |\mathcal{L}| \alpha_l \mathbf{H}^{(l)} \parallel |\mathcal{L}| \alpha_t \mathbf{H}^{(t)} \right]. \quad (27)$$

The factor $|\mathcal{L}|$ keeps the scale comparable to fixed fusion when the attention distribution is uniform.

The association predictor is trained by supervised imitation of low-energy reference decisions:

$$\mathcal{L}_{\text{ua}} = -\frac{1}{N} \sum_{i \in \mathcal{U}} \sum_{b \in \mathcal{B}} y_{i,b} \log P_{i,b}, \quad (28)$$

where $y_{i,b}$ is the reference association label. This loss trains the GNN to learn association preferences; feasibility and switch-off behavior are handled by the active-cell and repair stages below.

D. Active-BS Selection and QoS/Load Repair

Association probabilities alone do not determine which BSs should remain active, because a model may assign every UE while still keeping unnecessary BSs on. Therefore, the proposed framework uses a second-stage active-BS selector. For each BS b , a summary vector ψ_b is formed from probability statistics, demand-weighted probability mass, approximate load, channel quality, and distance statistics. The active score is

$$s_b = \sigma\left(\boldsymbol{\theta}^T \bar{\psi}_b + c\right), \quad (29)$$

where $\bar{\psi}_b$ is the standardized BS summary vector and $\sigma(\cdot)$ is the logistic sigmoid. The selected active set is the set of top-scoring BSs,

$$\hat{\mathcal{B}}_{\text{on}} = \text{Top}K_{b \in \mathcal{B}}(s_b). \quad (30)$$

The value of K is estimated from the average active-BS count of the reference solutions in the training set.

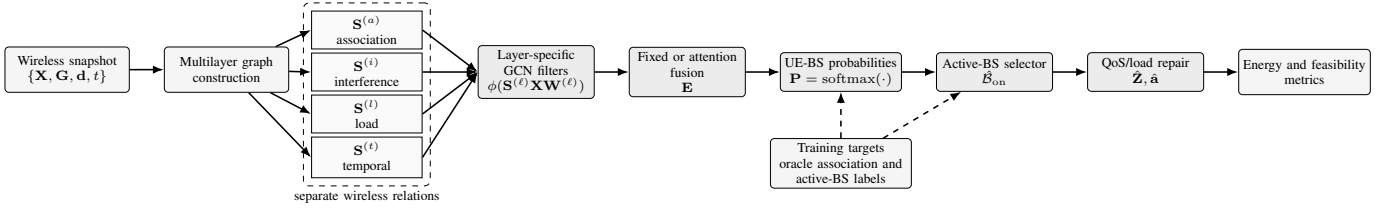


Fig. 1. Computational pipeline of the proposed framework. Wireless snapshots are converted into separate relation layers, processed by layer-specific GCN filters, fused into UE embeddings, and passed to association, active-BS selection, repair, and evaluation modules.

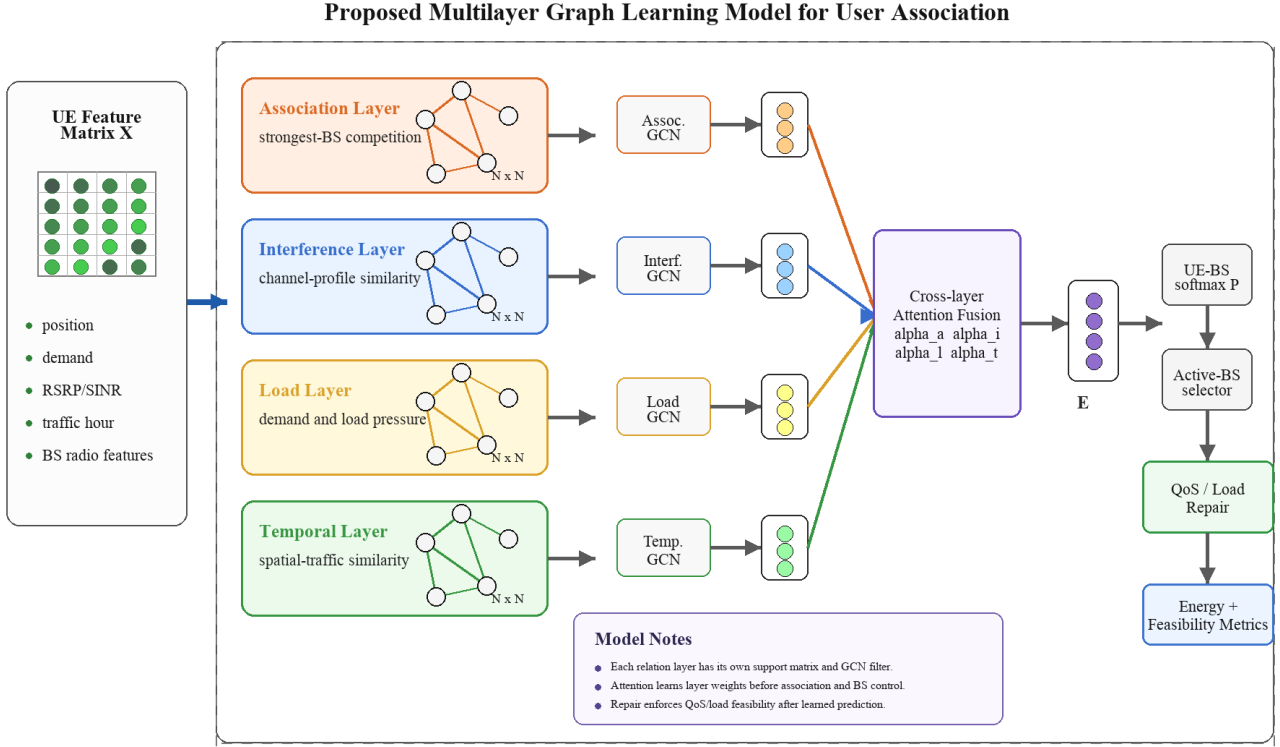


Fig. 2. Proposed multilayer graph-learning framework for energy-aware user association and BS switch-off. Separate wireless relation layers are preserved during graph propagation and fused before association, active-BS selection, and QoS/load repair.

The initial UE assignment chooses, for each UE, the most likely BS among $\hat{\mathcal{B}}_{\text{on}}$. A repair routine then checks the RSRP, rate, and load constraints in (7)–(11). If the selected active set cannot support the snapshot, additional BSs are activated according to their aggregate gain-demand score over stressed UEs. Algorithm 1 gives the complete inference procedure.

E. Mathematical Properties

Let $\mathfrak{M} = (\mathbf{X}, \{\mathbf{S}^{(\ell)}\}_{\ell \in \mathcal{L}})$ denote the multilayer input and define the layer-aware encoder F_{Θ} by (23)–(27). For a permutation matrix $\mathbf{\Pi}$, define the action

$$T_{\mathbf{\Pi}}(\mathfrak{M}) = (\mathbf{\Pi X}, \{\mathbf{\Pi S}^{(\ell)} \mathbf{\Pi}^T\}_{\ell \in \mathcal{L}}). \quad (31)$$

Define also the feasibility predicate

$$\Phi(\mathbf{Z}, \mathbf{a}) = 1 \iff (3), (4), (7), (8), (11) \text{ hold for } (\mathbf{Z}, \mathbf{a}). \quad (32)$$

Theorem 1 (Equivariance of the multilayer encoder). *For any permutation matrix $\mathbf{\Pi}$,*

$$F_{\Theta}(T_{\mathbf{\Pi}}(\mathfrak{M})) = \mathbf{\Pi} F_{\Theta}(\mathfrak{M}). \quad (33)$$

Proof. Let $\mathbf{H}^{(\ell)}$ and $\tilde{\mathbf{H}}^{(\ell)}$ be the layer- ℓ hidden matrices generated from $\mathfrak{M}$ and $T_{\mathbf{\Pi}}(\mathfrak{M})$, respectively. Then

$$\begin{aligned} \tilde{\mathbf{H}}^{(\ell)} &= \phi(\mathbf{\Pi S}^{(\ell)} \mathbf{\Pi}^T \mathbf{\Pi X W}^{(\ell)}) \\ &= \phi(\mathbf{\Pi S}^{(\ell)} \mathbf{X W}^{(\ell)}) = \mathbf{\Pi} \phi(\mathbf{S}^{(\ell)} \mathbf{X W}^{(\ell)}) \\ &= \mathbf{\Pi H}^{(\ell)}. \end{aligned} \quad (34)$$

Since concatenation and multiplication by scalar α_{ℓ} act on feature coordinates and not on UE indices,

$$\|_{\ell \in \mathcal{L}} \alpha_{\ell} \tilde{\mathbf{H}}^{(\ell)} = \mathbf{\Pi} \left(\|_{\ell \in \mathcal{L}} \alpha_{\ell} \mathbf{H}^{(\ell)} \right). \quad (35)$$

This proves (33). $\square$

Algorithm 1 Multilayer GNN Association With Active-BS Repair

Require: $(\mathbf{X}, \mathbf{G}, \mathbf{d}, t)$, parameters $\Theta = \{\mathbf{W}^{(\ell)}, \mathbf{W}_o, \mathbf{b}_o, \eta_\ell\}$, selector (θ, c) , K

Ensure: $\hat{\mathbf{Z}}, \hat{\mathbf{a}}$

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1:  $\mathcal{L} \leftarrow \{a, i, l, t\}$ 
2:  $\mathbf{S}^{(\ell)} \leftarrow \text{LayerSupport}_\ell(\mathbf{X}, \mathbf{G}, \mathbf{d}, t)$ ,  $\ell \in \mathcal{L}$ 
3: for  $\ell \in \mathcal{L}$  do
4:    $\mathbf{H}^{(\ell)} \leftarrow \phi(\mathbf{S}^{(\ell)} \mathbf{X} \mathbf{W}^{(\ell)})$ 
5: end for
6:  $\alpha_\ell \leftarrow \exp(\eta_\ell) / \sum_{r \in \mathcal{L}} \exp(\eta_r)$ 
7:  $\mathbf{E} \leftarrow \|\ell \in \mathcal{L}\| (|\mathcal{L}| \alpha_\ell \mathbf{H}^{(\ell)})$ 
8:  $\mathbf{P} \leftarrow \text{softmax}(\mathbf{E} \mathbf{W}_o + \mathbf{b}_o)$ 
9: for  $b \in \mathcal{B}$  do
10:   $\psi_b \leftarrow \text{BSFeatures}(b, \mathbf{P}, \mathbf{G}, \mathbf{d})$ 
11:   $s_b \leftarrow \sigma(\theta^T \psi_b + c)$ 
12: end for
13:  $\mathcal{A} \leftarrow \text{TopK}_{b \in \mathcal{B}}(s_b, K)$ 
14: repeat
15:   $z_{i,b} \leftarrow 0, \forall i, b$ 
16:  for  $i \in \mathcal{U}$  do
17:     $\mathcal{C}_i \leftarrow \{b \in \mathcal{A} : R_{i,b}(\mathcal{A}) \geq R_{\min}, P_{\text{tx}} g_{i,b} \geq \Gamma_{\text{rsrp}}\}$ 
18:    if  $\mathcal{C}_i = \emptyset$  then
19:       $\mathcal{C}_i \leftarrow \mathcal{A}$ 
20:    end if
21:     $b_i \leftarrow \arg \max_{b \in \mathcal{C}_i} P_{i,b}$ 
22:     $z_{i,b_i} \leftarrow 1$ 
23:  end for
24:   $\rho_b \leftarrow \sum_{i \in \mathcal{U}} z_{i,b} d_i / R_{i,b}(\mathcal{A})$ 
25:   $\mathcal{V} \leftarrow \{i : R_{i,b_i} < R_{\min} \text{ or } P_{\text{tx}} g_{i,b_i} < \Gamma_{\text{rsrp}} \text{ or } \rho_{b_i} > \rho_{\max}\}$ 
26:  if  $\mathcal{V} = \emptyset$  or  $|\mathcal{A}| = B$  then
27:    break
28:  end if
29:   $b^+ \leftarrow \arg \max_{b \in \mathcal{B} \setminus \mathcal{A}} \sum_{i \in \mathcal{V}} d_i g_{i,b}$ 
30:   $\mathcal{A} \leftarrow \mathcal{A} \cup \{b^+\}$ 
31: until  $\mathcal{V} = \emptyset$  or  $|\mathcal{A}| = B$ 
32:  $\hat{\mathbf{a}}_b \leftarrow \mathbf{1}\{b \in \mathcal{A}\}, \forall b$ 
33:  $\hat{z}_{i,b} \leftarrow z_{i,b}, \forall i, b$ 
34: return  $\hat{\mathbf{Z}}, \hat{\mathbf{a}}$ 

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Theorem 2 (Non-injectivity of uniform layer aggregation). *Let*

$$C(\mathfrak{M}) = (\mathbf{X}, \bar{\mathbf{S}}), \quad \bar{\mathbf{S}} = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \mathbf{S}^{(\ell)}, \quad (36)$$

be the uniform layer aggregation map. If $|\mathcal{L}| \geq 2$, then C is not injective. Hence any predictor $H(C(\mathfrak{M}))$ cannot distinguish all multilayer inputs.

Proof. It is sufficient to construct two distinct inputs with the same image under C . Consider two layers p and q , and let $\mathbf{A} \neq \mathbf{B}$. Define

$$\begin{aligned} \mathfrak{M}_1 : \quad & \mathbf{S}^{(p)} = \mathbf{A}, \quad \mathbf{S}^{(q)} = \mathbf{B}, \\ \mathfrak{M}_2 : \quad & \mathbf{S}^{(p)} = \mathbf{B}, \quad \mathbf{S}^{(q)} = \mathbf{A}, \end{aligned} \quad (37)$$

with all remaining layers and $\mathbf{X}$ identical. Then

$$C(\mathfrak{M}_1) = C(\mathfrak{M}_2), \quad \mathfrak{M}_1 \neq \mathfrak{M}_2. \quad (38)$$

Thus C is non-injective. Consequently, every predictor that factors through C assigns the same output to $\mathfrak{M}_1$ and $\mathfrak{M}_2$. In contrast, the layer-aware encoder contains the ordered terms $\phi(\mathbf{S}^{(p)} \mathbf{X} \mathbf{W}^{(p)})$ and $\phi(\mathbf{S}^{(q)} \mathbf{X} \mathbf{W}^{(q)})$; for $\mathbf{W}^{(p)} \neq \mathbf{W}^{(q)}$, the two constructed inputs need not have identical embeddings. $\square$

Lemma 1 (Attention normalization). *Let α be defined by (26). Then*

$$\alpha \in \Delta^{|\mathcal{L}|-1} = \{\mathbf{u} \in \mathbb{R}_+^{|\mathcal{L}|} : \sum_{\ell \in \mathcal{L}} u_\ell = 1\}. \quad (39)$$

Proof. For all ℓ , $\exp(\eta_\ell) > 0$; therefore $\alpha_\ell \geq 0$. Moreover,

$$\sum_{\ell \in \mathcal{L}} \alpha_\ell = \frac{\sum_{\ell \in \mathcal{L}} \exp(\eta_\ell)}{\sum_{r \in \mathcal{L}} \exp(\eta_r)} = 1. \quad (40)$$

$\square$

The layer-attention concentration reported in the experiments is measured by

$$H_\alpha = - \sum_{\ell \in \mathcal{L}} \alpha_\ell \log \alpha_\ell. \quad (41)$$

Theorem 3 (Finite monotone repair). *Let $\mathcal{A}^{(0)}$ be the active set initialized by (30). Algorithm 1 generates a sequence $\{\mathcal{A}^{(m)}\}_{m \geq 0}$ satisfying*

$$\mathcal{A}^{(0)} \subseteq \mathcal{A}^{(1)} \subseteq \dots \subseteq \mathcal{B}. \quad (42)$$

Moreover, the number of strict expansions is at most $B - |\mathcal{A}^{(0)}|$, and if the algorithm returns at iteration m with $\Phi(\hat{\mathbf{Z}}, \hat{\mathbf{a}}) = 1$, then $(\hat{\mathbf{Z}}, \hat{\mathbf{a}})$ is feasible for (15).

Proof. At any infeasible iteration, the update has the form

$$\mathcal{A}^{(m+1)} = \mathcal{A}^{(m)} \cup \{b_m^+\}, \quad b_m^+ \in \mathcal{B} \setminus \mathcal{A}^{(m)}. \quad (43)$$

Therefore (42) holds and each strict expansion increases $|\mathcal{A}^{(m)}|$ by one. Since $|\mathcal{B}| = B$, at most $B - |\mathcal{A}^{(0)}|$ strict expansions are possible. If the algorithm exits with $\Phi(\hat{\mathbf{Z}}, \hat{\mathbf{a}}) = 1$, then the association, activation, RSRP, rate, and load constraints hold by the definition of Φ . $\square$

Corollary 1 (Oracle bound). *Let $\mathcal{C}$ be the finite candidate set searched by the oracle and let $J(\mathbf{Z}, \mathbf{a}) = P_{\text{net}}(\mathbf{Z}, \mathbf{a})$. If*

$$(\mathbf{Z}^*, \mathbf{a}^*) \in \arg \min_{(\mathbf{Z}, \mathbf{a}) \in \mathcal{C}, \Phi(\mathbf{Z}, \mathbf{a})=1} J(\mathbf{Z}, \mathbf{a}), \quad (44)$$

then for every feasible $(\mathbf{Z}, \mathbf{a}) \in \mathcal{C}$,

$$J(\mathbf{Z}^*, \mathbf{a}^*) \leq J(\mathbf{Z}, \mathbf{a}). \quad (45)$$

Proof. The inequality is the defining property of the minimum over the feasible subset of $\mathcal{C}$. $\square$

TABLE I
BASELINE SIMULATION AND TRAINING CONFIGURATION

Parameter	Value
Number of BSs, B	7
Number of UEs, N	50
Service-region radius	650 m
Carrier frequency	3.5 GHz
Bandwidth	20 MHz
BS transmit power	20 W
Noise power	-94 dBm
Path-loss exponent	3.2
Shadowing standard deviation	6 dB
Minimum RSRP threshold	-115 dBm
Minimum rate, $R_{\min}$	1 Mbps
Traffic demand range	0.5-6 Mbps
Load limit, $\rho_{\max}$	1.0
BS fixed/dynamic/sleep power	90 W / 55 W / 8 W
Traffic periods, T	24
Training/test snapshots per seed	60 / 20
Random seeds	17, 23, 31
Hidden dimension	24
Training epochs / patience	80 / 20
Learning rate	0.018

V. EXPERIMENTAL DESIGN

A. Simulation Scenario

The experimental study uses simulated dense downlink cellular snapshots so that the true channel, traffic, association, load, and energy quantities are available for controlled comparison. Each snapshot contains B BSs deployed in a circular service region and N UEs drawn uniformly over the same region. One BS is placed at the center and the remaining BSs are placed on a ring, producing a dense small-cell layout with overlapping coverage. UE traffic demands vary across space and time: each snapshot is assigned a traffic period $t \in \{0, \dots, T-1\}$, and the demand distribution combines lognormal user variation, a periodic traffic factor, and a spatial hotspot.

The downlink channel gain $g_{i,b}$ includes distance-dependent path loss, lognormal shadowing, and small-scale fading. Unless otherwise stated, all BSs transmit with the same power and share the same bandwidth. The all-on network is used only as a reference operating point; learned and heuristic policies are allowed to switch off BSs through the active-BS vector $\mathbf{a}$ defined in Section III. Fig. 3 illustrates the simulated deployment and the information flow from snapshot inputs to radio-channel generation, multilayer graph construction, and control evaluation. Table I lists the baseline simulation configuration used for the initial comparative study.

B. Reference Decisions

Each training snapshot is labeled by a finite-search reference policy. The reference policy enumerates candidate active-BS subsets $\mathcal{A} \subseteq \mathcal{B}$ and assigns UEs greedily within each subset using channel quality, demand, and projected load. For each candidate, it computes an energy-penalized score

$$Q(\mathbf{Z}, \mathbf{a}) = P_{\text{net}}(\mathbf{Z}, \mathbf{a}) + \lambda_s (1 - \text{SR}(\mathbf{Z}, \mathbf{a})) + \lambda_\rho \left[\max_b \rho_b(\mathbf{Z}, \mathbf{a}) - \rho_{\max} \right]_+, \quad (46)$$

where $[x]_+ = \max(x, 0)$ and SR is the served ratio. The reference decision is the lowest-score feasible candidate in the searched set. This reference is not used as a claimed globally optimal solution of (15); it is used as a strong finite benchmark for supervised training and for measuring the energy gap of learned policies.

C. Compared Policies

The comparative study evaluates three groups of policies.

First, conventional association heuristics are included as non-learning baselines. The RSRP policy assigns each UE to the BS with maximum received power. The SINR policy assigns each UE to the BS with maximum all-on SINR. The load-aware policy sequentially assigns high-demand UEs to BSs that minimize projected load while maintaining radio feasibility when possible. These heuristics are repaired before evaluation using the same QoS/load guard applied to learned policies, so the comparison measures the quality of the association preference rather than failure of a postprocessing step.

Second, single-representation learning baselines are trained on the same reference labels. The flat MLP receives UE features only and does not use graph propagation. The aggregated GCN uses the average support

$$\bar{\mathbf{S}} = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \mathbf{S}^{(\ell)} \quad (47)$$

and therefore receives a single graph representation. These two baselines separate the value of graph propagation from the value of preserving separate relation layers.

Third, the proposed multilayer models are evaluated. The fixed-fusion ML-GCN uses (23)–(24), while the attention ML-GCN uses (26)–(27). Both models are combined with the active-BS selector and the QoS/load repair procedure in Algorithm 1. A finite-search reference policy is reported as an upper-bound benchmark within the searched candidate class.

D. Training Protocol

All learning models are trained by minimizing the association loss in (28) over the training snapshots. Node features are standardized using the training set statistics and the same transformation is applied to the test snapshots. The active-BS selector is trained after the association model by using the model probability outputs and the reference active sets. The number of selected active BSs is initialized from the rounded mean active-BS count of the reference decisions in the training set.

For each test snapshot, a learned model first outputs UE-to-BS probabilities, then active-BS scores, then an initial association over the selected active set. Algorithm 1 finally checks RSRP, rate, and load constraints and activates additional BSs only when the selected set cannot support the snapshot. All reported metrics are computed after this final repaired decision.

Dense Wireless Simulation Scenario

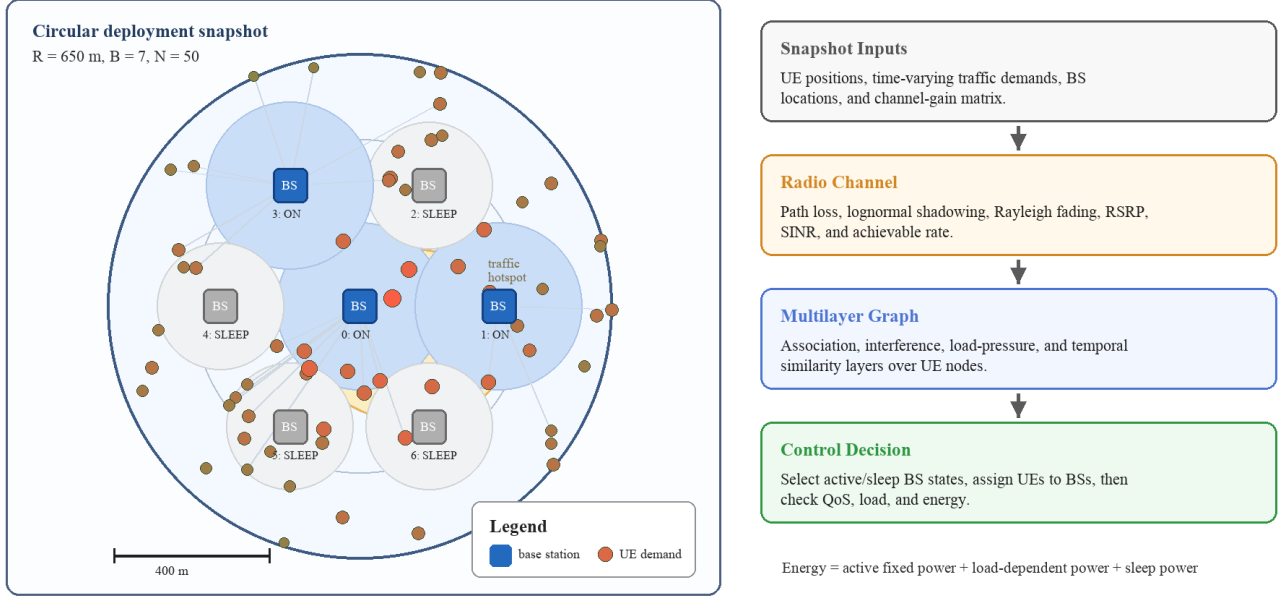


Fig. 3. Simulation scenario used for dense wireless user association and BS switch-off. Each snapshot contains a circular service region, center and ring BS deployment, randomly sampled UEs with time-varying traffic demand, channel gains generated from path loss, shadowing, and fading, and a control decision consisting of active/sleep BS states and UE-to-BS association.

E. Evaluation Metrics

The primary metric is network energy. Let P_{all} denote the energy consumed when all BSs are active under the same power model. The energy saving of a policy π is

$$\text{ES}_{\pi} = \frac{P_{\text{all}} - P_{\pi}}{P_{\text{all}}}, \quad (48)$$

where $P_{\pi} = P_{\text{net}}(\mathbf{Z}_{\pi}, \mathbf{a}_{\pi})$. The energy gap relative to the reference policy is

$$\text{Gap}_{\pi} = \frac{P_{\pi} - P_{\text{ref}}}{P_{\text{ref}}}, \quad (49)$$

where P_{ref} is the reference energy for the same snapshot.

Constraint satisfaction is measured by served ratio,

$$\text{SR}_{\pi} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{\left\{ P_{\text{tx}} g_{i,b_i} \geq \Gamma_{\text{rsrp}}, \right.} \\ \left. R_{i,b_i} \geq R_{\min}, \rho_{b_i} \leq \rho_{\max} \right\}. \quad (50)$$

Operational compactness and load behavior are reported using active-BS count

$$A_{\pi} = \sum_{b \in \mathcal{B}} a_{\pi,b}, \quad (51)$$

maximum load

$$\rho_{\max}^{\pi} = \max_{b \in \mathcal{B}} \rho_b, \quad (52)$$

and mean active load

$$\bar{\rho}_{\text{on}}^{\pi} = \frac{\sum_{b \in \mathcal{B}} a_{\pi,b} \rho_b}{\sum_{b \in \mathcal{B}} a_{\pi,b}}. \quad (53)$$

Assignment accuracy against reference labels is reported only as a diagnostic, because a policy can have low label agreement while still producing a feasible low-energy association.

F. Layer Ablation and Stress Tests

To evaluate the role of each relation layer, the experiments include single-layer and leave-one-layer-out variants of the ML-GCN. For a layer ℓ , the leave-one-layer-out energy effect is

$$\Delta_{\ell}^{(-)} = \text{ES}_{\text{full}} - \text{ES}_{-\ell}, \quad (54)$$

and the single-layer contribution is

$$\Delta_{\ell}^{(+)} = \text{ES}_{\ell} - \text{ES}_{\text{flat}}. \quad (55)$$

Positive $\Delta_{\ell}^{(-)}$ indicates that removing layer ℓ reduces energy saving, while positive $\Delta_{\ell}^{(+)}$ indicates that layer ℓ alone improves over a feature-only model.

Pairwise layer redundancy is measured before training using support-matrix dissimilarity,

$$D_{\ell,r} = \frac{\|\mathbf{S}^{(\ell)} - \mathbf{S}^{(r)}\|_F}{\|\mathbf{S}^{(\ell)}\|_F + \epsilon}, \quad \ell \neq r, \quad (56)$$

where $\epsilon > 0$ avoids division by zero. Attention concentration is measured using (41). In addition to the baseline setting in Table I, stress tests vary traffic intensity, UE density, BS density, and shadowing variance. These sweeps test whether energy savings remain meaningful when load and channel uncertainty change.

VI. RESULTS AND DISCUSSION

A. Overall Comparative Performance

Table II and Fig. 4 summarize the main multi-seed comparison. The finite-search reference achieves the lowest energy, with a mean energy saving of 63.2% relative to the all-on network and an average of 2.52 active BSs. The proposed ML-GCN is the strongest learned controller in the baseline setting: it obtains 55.2% energy saving, keeps the served ratio at 1.0, and uses 3.18 active BSs on average. Its mean energy gap relative to the finite-search reference is 23.2%.

The attention-weighted ML-GCN obtains a very similar result, with 54.7% energy saving and 3.23 active BSs. The aggregated GCN and flat MLP also learn cell switch-off behavior, but they remain slightly behind the layer-aware ML-GCN in energy and active-BS compactness. In contrast, the conventional RSRP, SINR, and load-aware policies keep all BSs active after repair and therefore save only about 17–18% energy. Their served ratio is close to one, but the energy gap exceeds 129% relative to the reference, indicating that local association rules alone do not identify compact active-cell configurations.

B. Representative Snapshot Behavior

Fig. 5 shows one representative test snapshot. The left panel gives the all-on SINR matrix, sorted by the reference association. The center and right panels compare the reference association and the ML-GCN association on the same UE/BS geometry. Both decisions preserve full service, but the learned policy uses one additional active BS in this snapshot. This is consistent with the aggregate results: the learned controller approaches the reference switch-off behavior but remains more conservative in order to satisfy QoS and load constraints after repair.

The snapshot also illustrates why assignment accuracy is not sufficient as a primary metric. A UE can be moved from one feasible BS to another and still produce a valid low-energy decision. Therefore, the correct evaluation target is not exact label reproduction; it is constrained network control measured by energy, active BS count, load, and served ratio.

C. Bandwidth and QoS Sensitivity

Fig. 6 evaluates bandwidth and QoS sensitivity. At 10 MHz, the network is highly resource constrained; all methods show low served ratio and energy saving is limited because most BSs must remain active. As bandwidth increases, the reference and learned policies exploit the additional capacity by switching off more BSs. At 20 MHz, ML-GCN reaches 51.5% energy saving with full served ratio, while RSRP and load-aware association remain near 17% because they do not actively compact the BS set. At 80 MHz, the learned controllers exceed 80% energy saving and approach the reference trend.

The QoS sweep is comparatively flat under the tested range $R_{\min} \in [0.5, 2]$ Mbps. This indicates that, in the baseline traffic and bandwidth setting, the dominant limiting factor is not the minimum-rate threshold itself but the load and active-cell structure created by the association policy.

This observation is useful for the final experimental design: more aggressive QoS stress should be combined with higher demand or lower bandwidth if rate feasibility is to become the dominant bottleneck.

To complement the one-dimensional sweeps, Fig. 7 reports a joint QoS-density stress surface for ML-GCN under three shadowing levels. The surface confirms that UE density is the dominant stressor in the tested regime: energy saving remains high when the network has 30–50 UEs, but decreases sharply as the UE count grows to 70–90. Changes in the QoS threshold produce a smaller effect than density, which is consistent with the earlier QoS sweep. This behavior indicates that the learned controller is mainly limited by load concentration and active-cell capacity under dense traffic, rather than by the minimum-rate threshold alone.

D. Scalability and Channel Robustness

Fig. 8 studies UE-density and shadowing sweeps. As the number of UEs increases from 30 to 90, all policies experience reduced energy saving because additional traffic requires more active BSs. The reference decreases from 73.8% to 39.2% energy saving, while ML-GCN decreases from 70.5% to 20.6%. The gap widens in the dense-load regime, which suggests that the current learned active-BS selector is conservative when many UEs compete for limited resources. Nevertheless, ML-GCN remains clearly above RSRP and load-aware association across the density sweep.

The shadowing sweep shows that the learned policies are stable under channel uncertainty in the tested range. ML-GCN maintains full served ratio and approximately 51–55% energy saving as the shadowing standard deviation varies from 2 to 14 dB. The reference energy saving also remains close to 62–63%. This robustness is expected because the multilayer representation uses multiple relational views: channel-profile similarity, association competition, demand/load structure, and temporal similarity. If one view becomes noisier, the remaining layers still provide useful structure for active-cell selection and association repair.

E. Layer Ablation

Fig. 9 reports layer ablation. Removing the load layer has the largest leave-one-layer-out penalty: the energy saving decreases from 55.4% to 53.2%, and the active-BS count increases from 3.2 to 3.4. This confirms that demand/load structure is important for deciding which BSs can remain active without overload. Removing the association, interference, or temporal layer also reduces energy saving, although with smaller penalties.

The single-layer comparison gives a complementary view. The interference layer alone gives the strongest single-layer improvement over the feature-only baseline, while association, load, and temporal layers are individually weaker but jointly useful. Taken together, these two ablation views support the central modeling claim: no single layer fully explains the energy-aware association decision. The multilayer model benefits from preserving heterogeneous dependencies before fusion.

TABLE II
MAIN MULTI-SEED PERFORMANCE SUMMARY

Policy	Energy saving	Reference gap	Served ratio	Active BSs	Max load
Reference	0.632 ± 0.013	0.000	1.000	2.52	0.831
ML-GCN	0.552 ± 0.022	0.232	1.000	3.18	0.827
Attn-ML-GCN	0.547 ± 0.025	0.242	1.000	3.23	0.821
Aggregated GCN	0.541 ± 0.010	0.259	1.000	3.30	0.819
Flat MLP	0.541 ± 0.019	0.259	1.000	3.27	0.832
Load-aware	0.171 ± 0.006	1.321	0.995	7.00	0.650
RSRP/SINR	0.180 ± 0.003	1.298	0.995	7.00	0.652

Main Comparative Performance Across Three Random Seeds

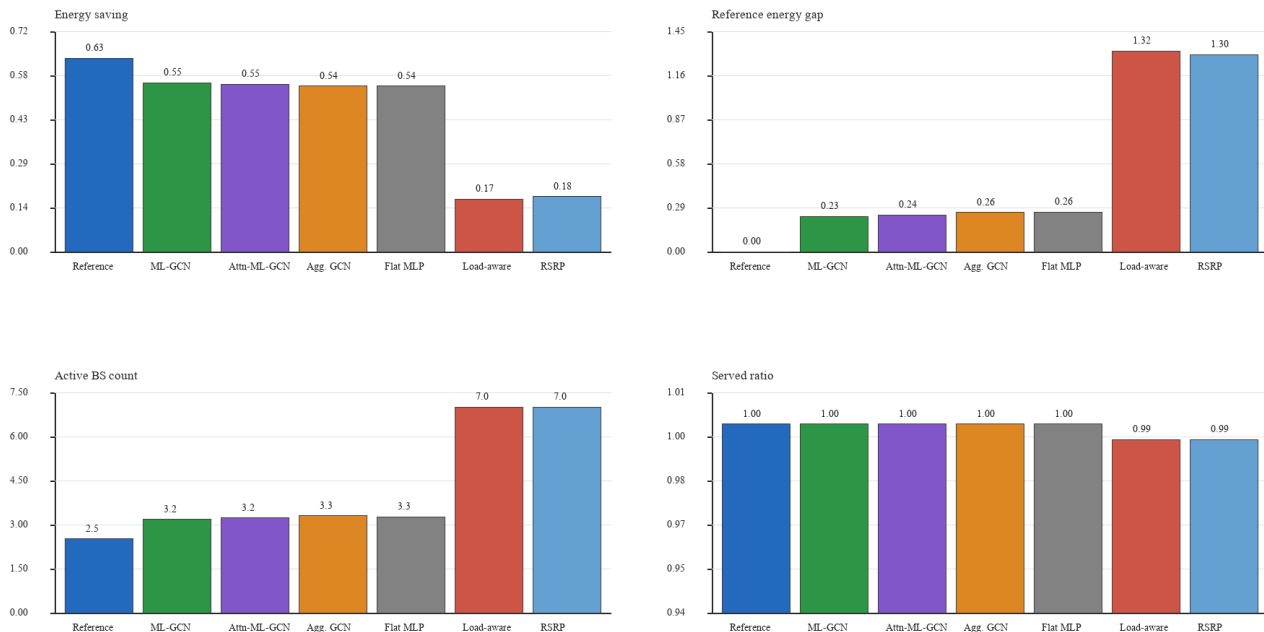


Fig. 4. Main comparative performance across three random seeds. Learned policies substantially reduce active BS count and energy consumption while preserving served ratio. The proposed ML-GCN gives the lowest mean energy among learned policies and the smallest reference gap.

F. Attention and Relation Dissimilarity

Fig. 10 shows the learned attention distribution and pairwise support-matrix dissimilarity. The attention model assigns comparable weights to the four layers, with a mild preference for interference and load. This explains why attention and fixed fusion perform similarly in the main comparison: the current global attention mechanism does not strongly suppress any layer. A more expressive snapshot-dependent attention mechanism may differentiate layers more sharply under heterogeneous traffic regimes.

The dissimilarity heatmap shows that the temporal layer is the most distinct from the association layer, while association and interference have moderate overlap. The nonzero off-diagonal dissimilarities support the use of separate relation layers rather than a single aggregated support. Even when attention weights are nearly uniform, the layer-specific filters still process different support matrices before fusion.

G. Runtime and Complexity

Table III reports a CPU timing benchmark and the corresponding online decision complexity. The benchmark uses the default network size, $N = 80$ UEs and $B = 7$ BSs, with 12 training snapshots and 5 test snapshots only for timing. The learned controllers are trained offline and then evaluated by measuring the median per-snapshot latency of probability prediction, active-BS selection, and QoS/load repair. Let L denote the number of relation layers, d the input feature dimension, h the hidden dimension, and $|E_\ell|$ the number of nonzero edges in layer ℓ .

The learned controllers add graph-processing cost relative to the simplest association rules, but their online latency remains in the same practical range for the default network size. The finite-search reference is much slower and scales exponentially with the number of BSs; it is therefore used as a bounded reference for evaluation rather than as an online controller. This runtime gap supports the role of ML-GCN as a deploy-

Representative Test Snapshot

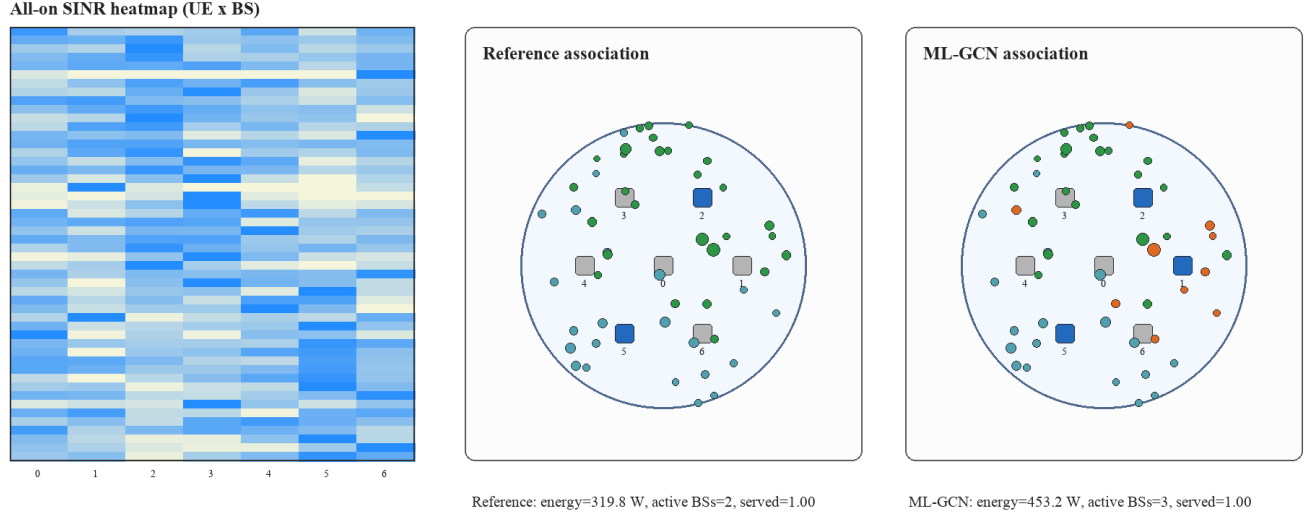


Fig. 5. Representative test snapshot. The heatmap shows all-on SINR over UE-BS pairs; the association maps show the finite-search reference and the ML-GCN decision after QoS/load repair. The learned policy maintains full service while selecting a compact active-BS set.

Bandwidth and QoS Sensitivity

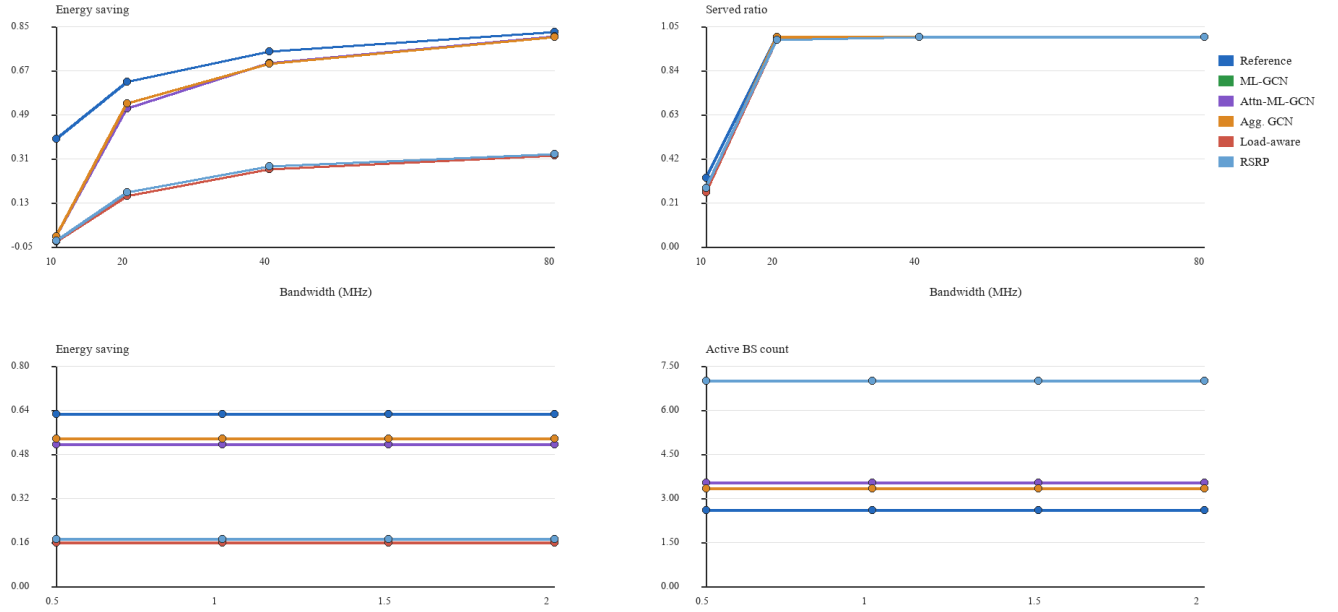


Fig. 6. Bandwidth and QoS sensitivity. Increasing bandwidth creates additional opportunity for BS switch-off, while the tested QoS range is not the dominant limiting factor under the baseline traffic load.

able approximation of the reference policy: it preserves the feasibility-driven control structure while avoiding exhaustive active-cell enumeration at inference time.

H. Discussion

The results show three main findings. First, energy-aware user association is primarily an active-cell control problem,

not only an assignment problem. RSRP, SINR, and load-aware association maintain high service quality but fail to produce compact active-BS sets, which leaves large energy savings unused. Second, supervised graph learning with feasibility repair can learn the switch-off behavior of the reference policy while preserving QoS and load constraints. Third, preserving layer-specific wireless relations provides a measurable advantage

Joint QoS-Density Surface for ML-GCN Energy Saving

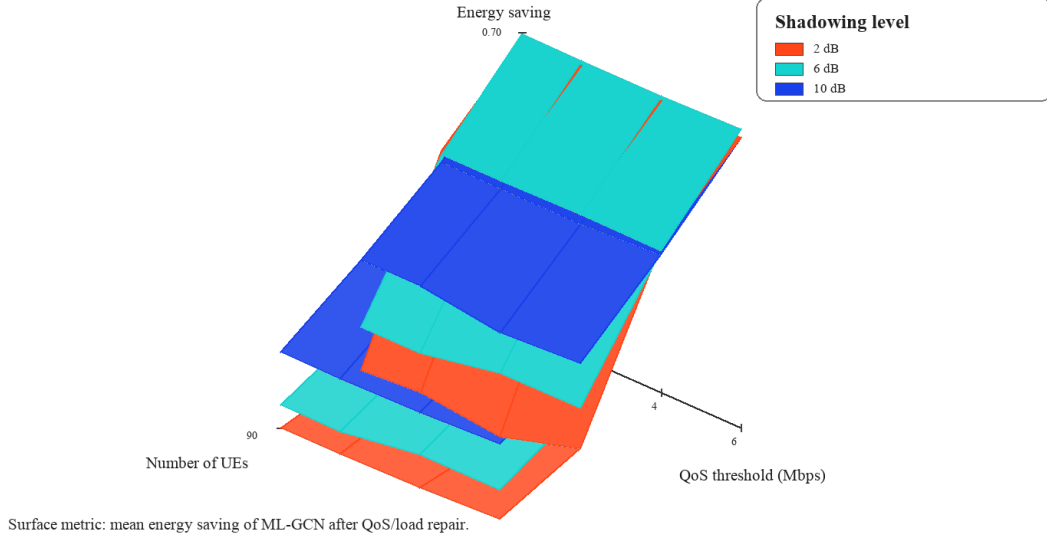


Fig. 7. Joint QoS-density surface for the proposed ML-GCN. Each surface reports mean energy saving after QoS/load repair for a different shadowing standard deviation. The figure shows that UE density has a stronger impact on energy saving than the tested QoS-threshold variation.

Scalability and Channel-Robustness Sweeps

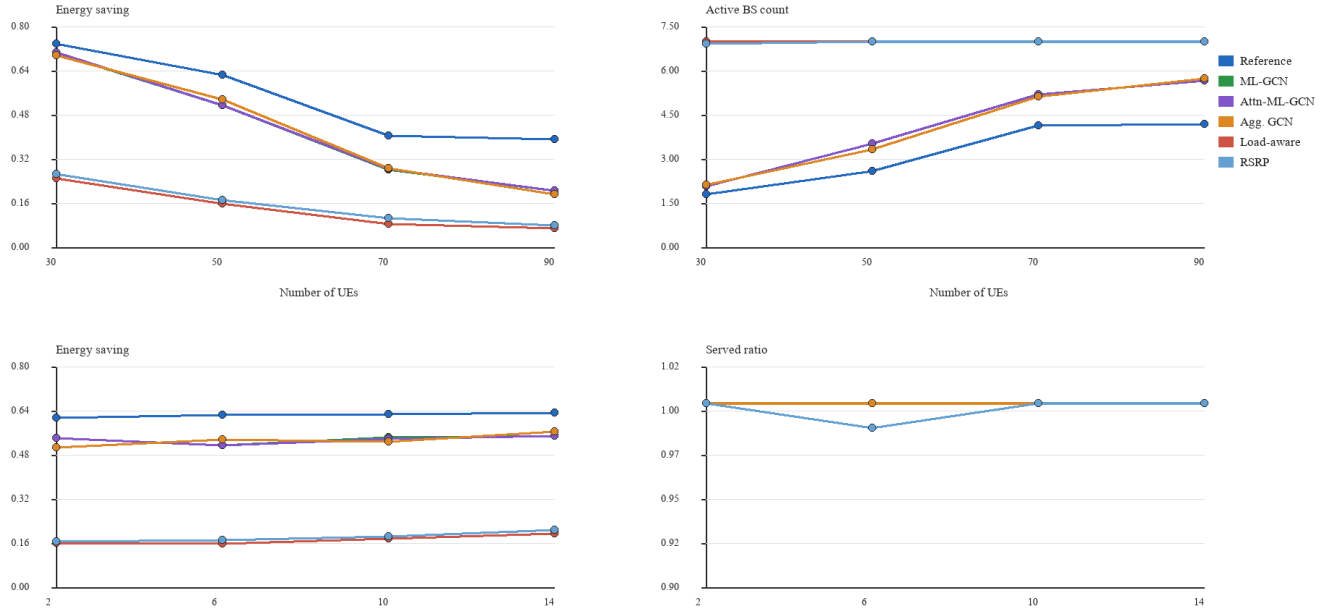


Fig. 8. Scalability and channel-robustness sweeps. Higher UE density reduces energy-saving opportunity and increases active BS count. Under the tested shadowing levels, the learned policies preserve service feasibility and remain substantially more energy-efficient than conventional association rules.

over graph aggregation, particularly in active-BS compactness and reference energy gap.

The present attention model should be interpreted cautiously. Its global attention weights are useful for inter-

Layer Ablation and Single-Layer Contribution

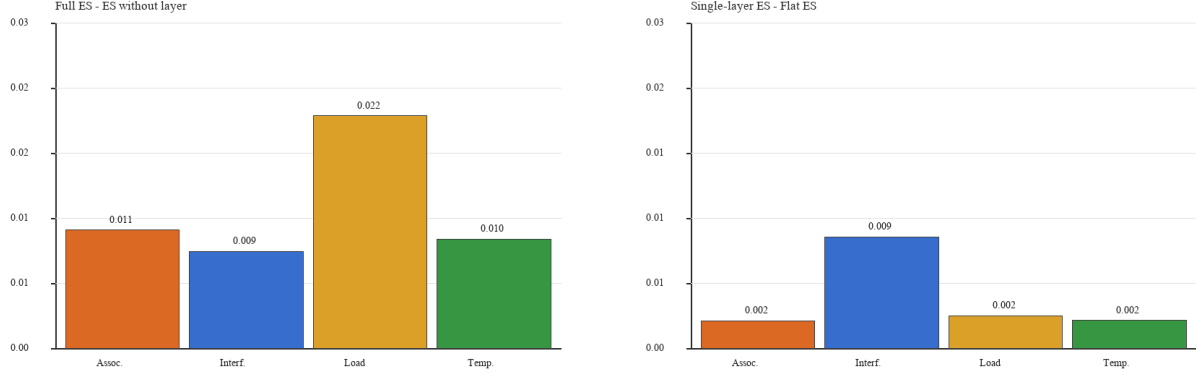


Fig. 9. Layer ablation analysis. The load layer has the largest leave-one-layer-out effect, while the interference layer gives the strongest single-layer gain over the feature-only baseline.

Layer Attention and Relation Dissimilarity

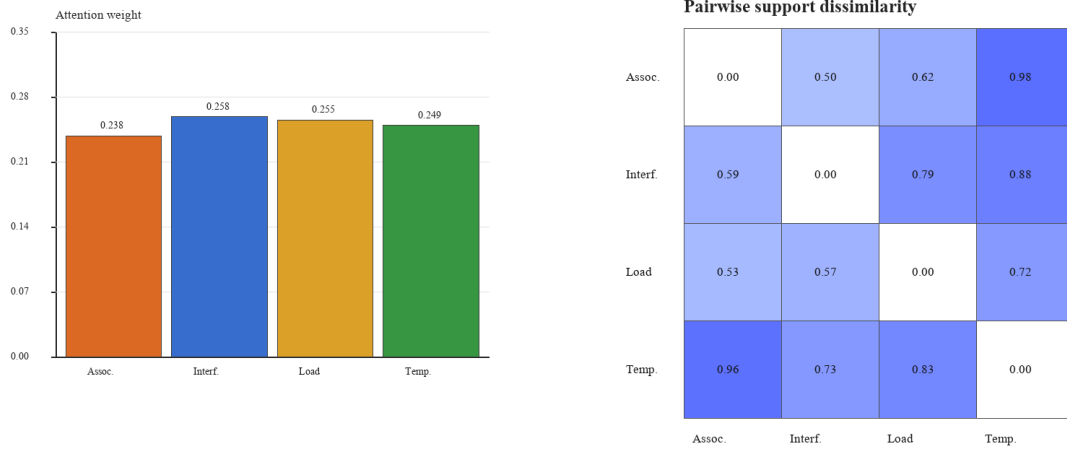


Fig. 10. Layer attention and relation dissimilarity. Attention weights are nearly balanced across relation layers, while the support matrices remain structurally distinct.

TABLE III
RUNTIME AND ONLINE COMPLEXITY BENCHMARK

Policy	Offline fit (s)	Median decision latency (ms)	Online decision complexity
RSRP	—	18.8	$\mathcal{O}(NB)$
SINR	—	12.6	$\mathcal{O}(NB)$
Load-aware	—	30.0	$\mathcal{O}(NB + N \log N)$
Flat MLP	0.26	22.9	$\mathcal{O}(Ndh + Nh^2 + NhB)$
Aggregated GCN	0.30	26.1	$\mathcal{O}(E d + Ndh + NhB)$
ML-GCN	0.39	23.6	$\mathcal{O}(\sum_{\ell=1}^L E_{\ell} d + LNd h + LNhB)$
Attn-ML-GCN	0.38	26.9	$\mathcal{O}(\sum_{\ell=1}^L E_{\ell} d + LNd h + LNhB)$
Reference search	—	837.5	$\mathcal{O}(2^B NB)$

pretability, but they do not yet produce a large performance separation from fixed fusion. This is not a weakness of the

multilayer formulation; rather, it indicates that the current attention parameterization is conservative. Future experiments can use snapshot-dependent attention, stress-specific training, or explicit layer-diversity regularization to make the attention mechanism more adaptive.

Finally, the finite-search reference is not a practical online controller for large networks because it evaluates many active-BS candidates. Its role is to provide a strong benchmark and supervised signal. The learned ML-GCN replaces this search with a fast feed-forward policy and a finite monotone repair step, thereby providing a scalable approximation to energy-aware association under explicit feasibility constraints.

VII. CONCLUSION

This paper studied energy-aware user association in wireless access networks using a multilayer graph-learning formulation. The proposed framework models a network snapshot through heterogeneous UE-UE relation layers that encode association competition, interference similarity, load coupling, and temporal traffic structure. These layers are processed by an ML-GCN policy and followed by an explicit QoS/load repair step, so that the learned association decision is evaluated as a constrained network-control action rather than as a pure classification task.

The simulation results show that layer-aware graph learning can substantially reduce network energy consumption while maintaining service feasibility. In the baseline setting, the fixed-fusion ML-GCN achieved 55.2% mean energy saving relative to the all-on network, maintained a served ratio of 1.0, and used 3.18 active BSs on average. The finite-search reference achieved 63.2% energy saving, whereas conventional RSRP, SINR, and load-aware policies saved only about 17–18% because they did not explicitly compact the active-BS set. Sensitivity studies further showed that the proposed method remains effective across bandwidth, QoS-threshold, UE-density, and shadowing variations. The ablation analysis confirmed that all relation layers contribute to the learned control decision, with the load layer producing the largest leave-one-layer-out penalty and the interference layer providing the strongest single-layer gain. The runtime benchmark also showed that ML-GCN avoids the exponential active-BS enumeration cost of the reference search and provides a feed-forward decision mechanism with practical online latency for the simulated network size.

The study has several limitations. First, the evaluation is based on a simulation environment with controlled propagation, traffic, and power-model assumptions; although these assumptions allow reproducible comparative analysis, they do not capture all operational effects of a deployed cellular network. Second, the finite-search reference is tractable only for the considered BS scale and is used as a bounded benchmark rather than a scalable optimal solver. Third, the current attention mechanism uses global layer weights, which limits its ability to adapt relation importance on a snapshot-by-snapshot basis. Finally, the repair step enforces feasibility after learning, but it is not yet jointly optimized with the ML-GCN objective.

Future work will extend the framework in four directions. The first direction is validation under larger and more realistic network scenarios, including irregular BS deployments,

mobility traces, nonuniform traffic hotspots, and multi-cell interference coordination. The second direction is the development of adaptive attention and layer-selection mechanisms that can change relation importance according to traffic load, channel uncertainty, and topology. The third direction is to integrate feasibility constraints more directly into the learning objective through constrained learning, differentiable repair, or safe reinforcement learning. The fourth direction is to evaluate transfer across network sizes and operating regimes, so that a model trained on one deployment can generalize to different densities, bandwidths, and service requirements. These extensions would move the proposed multilayer graph framework closer to practical energy-efficient control in future intelligent wireless networks.

DATA AND CODE AVAILABILITY

The source code, simulation configurations, generated result files, MATLAB figure scripts, and manuscript assets used in this study are publicly available at <https://github.com/kazim01011/multilayer-ua-energy>. The repository contains the simulator, multilayer graph construction routines, learning models, baseline policies, reproducibility scripts, and aggregate CSV outputs used to generate the reported tables and figures.

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